

Criterio de la raíz

Todas las que se pueden resolver por criterio de la raíz se pueden resolver por el criterio de la razón

Criterio de la raíz

La serie $\sum_{n=p}^{\infty} (a_n)^n \leftarrow$ forma

No p

Converge si
 $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1$

Diverge si
 $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} > 1$ o ∞

$\lim_{n \rightarrow \infty} \sqrt[n]{k} = 1$, k constante

$\lim_{n \rightarrow \infty} \sqrt[n]{k} = 1$, k una constante

$\lim_{n \rightarrow \infty} \sqrt[n]{P(k)} = 1$, $P(k)$ un polinomio

$\lim_{n \rightarrow \infty} \sqrt[n]{\frac{P(k)}{Q(k)}} = 1$, $P(k)$, $Q(k)$ polinomios

$$\sqrt[n]{n^n} = n$$

$$\infty$$

$$\sum_{n=1}^{\infty} \left(\frac{3n^2 + 3n + 1}{5n^2 + 5} \right)^{3n}$$

$$\lim_{n \rightarrow \infty} n \sqrt[n]{\left(\frac{3n^2 + 3n + 1}{5n^2 + 5} \right)^{3n}}$$

$$\lim_{n \rightarrow \infty} n \sqrt[n]{\left[\left(\frac{3n^2 + 3n + 1}{5n^2 + 5} \right)^3 \right]^n}$$

$$\lim_{n \rightarrow \infty} \left(\frac{3n^2 + 3n + 1}{5n^2 + 5} \right)^3$$

$$\lim_{n \rightarrow \infty} \left(\frac{3}{5} \right)^3 = \left(\frac{3}{5} \right)^3 = \frac{3^3}{5^3} = \frac{9}{25} < 1$$

$$(x^2)^n = x^{2n}$$

$$x^{2n} = (x^n)^2 = (x^2)^n$$

$$\sqrt[n]{x \cdot y} = \sqrt[n]{x} \cdot \sqrt[n]{y}$$

$$\sqrt[n]{n^n} = n$$

ist Divergente

∞ $\sum$

$$n=1 \quad \frac{n^n}{2^{n^2}}$$

$$(a_n)^n$$

$$\lim_{n \rightarrow \infty} \frac{n \sqrt[n]{n^n}}{2^{n^2}}$$

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n} \sqrt[n]{\cancel{n^n}}}{n \sqrt{2^{n^2}}}$$

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt[n]{2^{n^2}}}$$

$$\sqrt[a]{\frac{a}{b}} = \frac{a}{b}$$

$$(x^a)^b = x^{ab} = x^{ba} = (x^b)^a$$

$$\lim_{n \rightarrow \infty} \frac{n}{\cancel{n^2} 2^{\cancel{n}}}$$

$$\lim_{n \rightarrow \infty} \frac{n}{2^n}$$

$$2^p < a^n$$

$$= 0$$

CADENA DE TÉRMINOS DOMINANTES

Sean $k \in \mathbb{R}, a, p \in \mathbb{R}^+, a > 1$ entonces para n suficientemente grande se tiene que:

$$k \ll \ln n \ll n^p \ll a^n \ll n! \ll n^n$$

$$5. \sum_{k=2}^{\infty} \frac{k^5 \cdot (k^3 + 2)^{k+1}}{(2k+1)^{3k}}$$

$$n = k$$

$$\sqrt{a \cdot b} = \sqrt{a} \cdot \sqrt{b}$$

$$\lim_{n \rightarrow +\infty} \frac{n \sqrt[n]{n^5 (n^3 + 2)^{n+1}}}{(2n+1)^{3n}}$$

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

$$\lim_{n \rightarrow +\infty} \frac{\cancel{n} \sqrt[n]{n^5} \cdot \sqrt[n]{(n^3 + 2)^{n+1}}}{\sqrt[n]{(2n+1)^{3n}}}$$

$$\sqrt[n]{n^p} = 1$$

$$\lim_{n \rightarrow +\infty} \frac{\sqrt[n]{(n^3 + 2)^{n+1}}}{\sqrt[n]{(2n+1)^{3n}}}$$

$$x^{a+b} = x^a \cdot x^b$$

$$\lim_{n \rightarrow +\infty} \frac{n \sqrt[n]{(n^3 + 2)^n (n+2)^1}}{\sqrt[n]{(2n+1)^{3n}}}$$

$$\sqrt[n]{x^b} = x^{\frac{b}{n}}$$

$$\lim_{n \rightarrow +\infty} \frac{n \sqrt[n]{(n^3 + 2)^n (n+2)^1}}{(2n+1)^3}$$

$$\sqrt[n]{2^1} = 2^{\frac{1}{n}}$$

$$(2n+1)^3$$

$$\lim_{n \rightarrow +\infty} n \sqrt[n]{e(n)} = 1$$

$$\lim_{n \rightarrow +\infty} \frac{\cancel{n} \sqrt[n]{(n^3 + 2)^n} \cdot \sqrt[n]{(n+2)^1}}{(2n+1)^3}$$

CADENA DE TÉRMINOS DOMINANTES

Sean $k \in \mathbb{R}, a, p \in \mathbb{R}^+, a > 1$ entonces para n suficientemente grande se tiene que:

$$k \ll \ln n \ll n^p \ll a^n \ll n! \ll n^n$$

$$\lim_{n \rightarrow +\infty} \frac{n^3}{(2n+3)^3}$$

$$n + \overset{0}{\cancel{n}} = n$$

$$\cancel{1} \cdot n = n$$

$$\lim_{n \rightarrow +\infty} \frac{n^3}{(2n)^3} = \frac{\cancel{n^3}}{8\cancel{n^3}} = \frac{1}{8} < 1$$

Converge

$$\sum_{n=1}^{\infty} \frac{(-1)^n \cdot \left(2 + \frac{1}{n}\right)^{2n}}{e^n}$$

$$\lim_{n \rightarrow +\infty} \left| \frac{\sqrt[n]{(-1)^n} \cdot \sqrt[n]{\left(2 + \frac{1}{n}\right)^{2n}}}{\sqrt[n]{e^n}} \right|$$

$$\lim_{n \rightarrow +\infty} \frac{1 \cdot \left(2 + \frac{1}{n}\right)^2}{e} \quad \frac{1}{+\infty} = 0$$

$$\frac{4}{e} \approx 1.4715$$

∴ Es Divergente

$$\textcircled{b} \sum_{n=1}^{\infty} \frac{3+5n^2}{(1+2n)^n}$$

$$\sqrt[n]{|a_n|}^n = |a_n|$$

$$\lim_{n \rightarrow +\infty} \sqrt[n]{\left| \frac{3+5n^2}{(1+2n)^n} \right|}$$

Recordar valor absoluto | |

$$\lim_{n \rightarrow +\infty} \frac{\sqrt[n]{3+5n^2}}{\sqrt[n]{(1+2n)^n}}$$

$$\sqrt[n]{|a_n|} = 1$$

$$\lim_{n \rightarrow +\infty} \frac{1}{1+2n} = 0 < 1$$

∴ Converge

$$d) \sum_{n=1}^{\infty} \left(\frac{-n+5}{2n+9} \right)^{2n+1}$$

$$\lim_{n \rightarrow +\infty} \sqrt[n]{\left| \left(\frac{-n+5}{2n+9} \right)^{2n+1} \right|} \quad x^{a+b} = x^a \cdot x^b$$

$$\lim_{n \rightarrow +\infty} \sqrt[n]{\left(\frac{n+5}{2n+9} \right)^{2n}} \cdot \sqrt[n]{\left(\frac{-n+5}{2n+9} \right)}$$

se borra el - por el | | 1

$$\lim_{n \rightarrow +\infty} \sqrt[n]{\left[\left(\frac{n+5}{2n+9} \right)^2 \right]^n}$$

$$\lim_{n \rightarrow +\infty} \left(\frac{n+5}{2n+9} \right)^2$$

$$\lim_{n \rightarrow +\infty} \left(\frac{n}{2n} \right)^2 = \frac{1}{4} < 1$$

∴ Converge

Series de potencias

$$\sum_{n=p}^{\infty} a_n (x-c)^n$$

$$a_n (x-c)^n$$

Tiene 2 letras
normalmente x y n
 $\forall x \in \mathbb{R}$

Razon
Raz NADA
MAS

L = El resultado del $\lim$

$n \rightarrow \infty$

del criterio de la razon o
del criterio de la raiz

Caso 1: $L = K$, K constante $\neq 0$

$$L |x-K| < 1$$

Siempre poner
 < 1 por condici3n
de convergencia de
la razon $\vee$ raiz

$$|x-K| < \frac{1}{L}$$

$$-\frac{1}{L} < x-K < \frac{1}{L}$$

K puede
ser 0
 $|x|$

$$K - \frac{1}{L} < x < K + \frac{1}{L}$$

$$\left] K - \frac{1}{L}, K + \frac{1}{L} \right[$$

Intervalo de convergencia

Radio (R)

$$R = \frac{1}{L}$$

Caso 2: $L=0$

Números reales

Intervalo de convergencia = $\mathbb{R}$

Radio = $+\infty$

Caso 3: $L=+\infty$

Intervalo = $x=\pi$, $x-\pi=0$
 $x=\pi$

Radio = 0

Caso 3: $L = K$, K constante $\neq 0$

∞

$$\sum_{n=0}^{\infty} (x-3)^n$$

$$n \rightarrow 0$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|x-3|}$$

$$\lim_{n \rightarrow \infty} |x-3|$$

$|x-3|$ $\lim_{n \rightarrow \infty} 1$ $\rightarrow$ el x sale afuera
con todo lo que tenga
 $|x-3|$

$$I. |x-3| < 1$$

$$|x-3| < 1$$

$$-1 < x-3 < 1$$

$$3-1 < x < 3+1$$

$$2 < x < 4$$

$$I = 32, 72$$

$$R = \frac{3}{2} = \frac{3}{2} = 1$$

Caso 2: $L=0$ $I=\mathbb{R}$ $\text{radio} = +\infty$

$$\sum_{n=1}^{\infty} \frac{h(x+3)^n}{3 \cdot 5 \cdot 7 \cdots (2n+1)}$$

Con criterio de la razón

$$\lim_{n \rightarrow +\infty} \frac{(n+1) \cancel{(x+3)^{n+1}}}{\cancel{3 \cdot 5 \cdot 7 \cdots (2n+1)} (2n+3) \cancel{h(x+3)^n}} = \frac{n+1}{2n+3} \cdot \frac{1}{h}$$

$$\lim_{n \rightarrow +\infty} \frac{(n+1)(x+3)^{\cancel{n+1}}}{(2n+3) \cancel{h(x+3)^n}} = \frac{n+1}{2n+3} \cdot \frac{1}{h}$$

$$\lim_{n \rightarrow +\infty} \frac{(n+1)(x+3)}{2n^2 + 3n}$$

$$|x+3| \lim_{n \rightarrow +\infty} \frac{n+1}{2n^2 + 3n}$$

Sale A Fuera
Con valor
absoluto

$$|x+3| \lim_{n \rightarrow +\infty} \frac{\cancel{n}}{2\cancel{n^2}} = \frac{1}{2h} = 0$$

$$I = \mathbb{R} \quad \text{radio} = +\infty$$

Caso 3: $L = +\infty$

$$\sum_{n=0}^{\infty} n^n (1-x)^n, \quad \text{usando el crit. de raíz}$$

$$\lim_{n \rightarrow +\infty} \sqrt[n]{n^n} \sqrt[n]{(1-x)^n}$$

$$\lim_{n \rightarrow +\infty} n |1-x|$$

$$|1-x| \lim_{n \rightarrow +\infty} n = +\infty$$

$\hookrightarrow |1-x| > 1$

$$\begin{aligned} 1-x &= 0 \\ -x &= -1 \\ x &= 1 \end{aligned}$$

$$\boxed{I = \{1\} \quad \text{radio} = 0}$$

Ejemplos mixtos

2. $\sum_{n=0}^{\infty} (4x+6)^n$ por raíz

$$\lim_{n \rightarrow \infty} \sqrt[n]{(4x+6)^n} \quad L = K \neq 0$$

$$|4x+6| \lim_{n \rightarrow \infty} 1 = 1 \quad \leftarrow L$$

$$1 \cdot |4x+6| < 1$$

$$|4x+6| < 1$$

$$-1 < 4x+6 < 1$$

$$-6-1 < 4x < 1-6$$

$$-7 < 4x < -5$$

$$\frac{-7}{4} < x < \frac{-5}{4}$$

$$I = \left] \frac{-7}{4}, \frac{-5}{4} \right[$$

$$8. \sum_{n=0}^{\infty} \frac{(x+1)^n}{n+1}$$

$$\lim_{h \rightarrow \infty} \frac{\sqrt[n]{(x+1)^h}}{\sqrt[n]{n+1}} = 1$$

$$\sqrt[n]{n} = 1$$

$$|x+1| \lim_{h \rightarrow \infty} \frac{1}{1} = 1$$

$$1 - |x+1| < 1$$

$$|x+1| < 1$$

$$-1 < x+1 < 1$$

$$-2 < x < 0$$

$$I =]-2, 0[$$

Pregunta #7: Determine el intervalo de convergencia de la siguiente serie de potencias

$$\sum_{n=1}^{\infty} \frac{n! (x-1)^n}{3 \cdot 5 \cdot 7 \cdot \dots \cdot (2n+1)}$$

$$\lim_{n \rightarrow \infty} \frac{(n+1)! \cancel{(x-1)^n} (x-1)^2}{\cancel{3 \cdot 5 \cdot 7 \cdot \dots \cdot (2n+1)} (2n+3) \cancel{n! (x-1)^n}}$$

$$\lim_{n \rightarrow \infty} \frac{(n+1) \cancel{n!} |x-1|}{(2n+3) \cancel{n!}}$$

$$|x-1| \lim_{n \rightarrow \infty} \frac{n+1}{2n+3}$$

$$|x-1| \lim_{n \rightarrow \infty} \frac{\cancel{n}}{\cancel{2n}} = \frac{1}{2} < 1$$

$$\frac{1}{2} |x-1| < 1$$

$$|x-1| < 2$$

$$-2 < x-1 < 2$$

$$-1 < x < 3$$

$$I =]-1, 3[$$

Series alternadas

$$\sum_{h=p}^{\infty} a_h \underbrace{(-I)^h}_{\rightarrow (-I)^{h-1}, (-I)^{h+1}}$$

Para que converja

1) a_n sea decreciente

$\vdash f'(x)$
 $\vdash \frac{a_{n+1}}{a_n} \geq 1, \quad \begin{matrix} v \rightarrow \text{crece} \\ x \rightarrow \text{decrece} \end{matrix}$

$$2) \lim_{n \rightarrow +\infty} a_n = 0$$

Convergencia absoluta y condicional

$$\sum_{h=p}^{\infty} a_n \cdot (-1)^n \rightarrow \sum_{h=p}^{\infty} a_n \quad \text{Si } \gamma \text{ converge}$$

Convergenza assoluta

$$\sum_{n=p}^{\infty} a_n \cdot (-1)^n \rightarrow \sum_{n=p}^{\infty} a_n \quad \text{Diverge entonces aplicamos el criterio}$$

(Decreciente y $\lim_{n \rightarrow \infty} a_n = 0$)

Si se cumple el criterio

Converge Condicionalmente

∞

$$\sum_{n=2}^{\infty} \frac{1}{n^2} \cdot (-1)^n$$

Convergence absolutely?

 ∞

$$\sum_{n=2}^{\infty} \frac{1}{n^2}, p, p > 1, \therefore \text{Converge}$$

 ∞

$$\therefore \sum_{n=2}^{\infty} \frac{1}{n^2} \cdot (-1)^n$$

Converge

Absolutely

 ∞

$$\sum_{n=2}^{\infty} \frac{1}{n} \cdot (-1)^n$$

Convergence absolutely?

 ∞

$$\sum_{n=2}^{\infty} \frac{1}{n}, p, p \leq 1, \text{Diverge}$$

Converge conditionally?

$$\text{Decrease? } f'(x) = \frac{-1}{x^2} \checkmark \quad a_n \rightarrow 0? \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \checkmark$$

$\therefore$ Converge conditionally

∞

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n \ln(n)}$$

 ∞

$$\sum_{n=1}^{\infty} \frac{1}{n \ln(n)} (-1)^{n-1}$$

 ∞

$$\sum_{n=1}^{\infty} \frac{1}{n \ln(n)}$$

Converge Absolutamente?
Por crit de la integral

$$f(x) = \frac{1}{x \ln(x)}$$

$$f'(x) = \frac{-1}{x^2 \ln^2(x)} \cdot \ln(x) + \frac{1}{x^2}$$

$$= -\frac{\ln(x) + 1}{x^2 \ln^2(x)} \quad \checkmark \quad \text{Como es negativa decrece}$$

$$\int_2^{\infty} \frac{1}{x \ln(x)} \quad u = \ln(x) \quad du = \frac{1}{x}$$

$$\int \frac{1}{u} du \rightarrow \ln|u| \quad \begin{matrix} \ln|u| & \xrightarrow{+\infty} \\ \ln|\ln|x|| & \downarrow \\ \ln|+\infty| - \ln|\ln|2|| & \downarrow \\ +\infty & \text{v! Diverge} \end{matrix}$$

Convergença condicionalmente?

$$f \sim \frac{-\ln(x) + 1}{x^2 \ln^2(x)} \quad \checkmark \quad \text{Como es negativa decorece}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n \ln(n)} = 0$$

$$\therefore \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n \ln(n)}$$

Convergença condicionalmente

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1} 2^n \cdot n!}{5 \cdot 8 \cdot 11 \dots (3n+2)}$$

Converge Absolutamente?

$$\sum_{n=1}^{\infty} \frac{2^n \cdot n!}{5 \cdot 8 \cdot 11 \dots (3n+2)}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{2^n \cdot 2 \cdot (n+1) \cdot n!}}{\cancel{5 \cdot 8 \cdot 11 \dots (3n+2)} (3n+5)} \cdot \frac{\cancel{2^n \cdot n!}}{\cancel{5 \cdot 8 \cdot 11 \dots (3n+2)}}$$

$$\lim_{n \rightarrow \infty} \frac{2n+2}{3n+5} = \lim_{n \rightarrow \infty} \frac{2}{3} = \frac{2}{3} < 1$$

$\therefore$ Converge Absolutamente